

A parabolic counterexample to the Gaussian Neumann-to-Dirichlet condition

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Status

Candidate counterexample; likely valid, pending human review.

Source Question

The source is Alexander V. Kolesnikov and Emanuel Milman, *Sharp Poincaré-type inequality for the Gaussian measure on the boundary of convex sets*, arXiv:1601.02925v2 [1]. Question 3.4 on page 11 asks whether condition (3.2) holds for standard Gaussian measure with

$$F(v) = \frac{1}{v} - (\log I_\gamma)'(v), \quad I_\gamma(v) = \varphi(\Phi^{-1}(v)).$$

Here is the question in the source.

Unfortunately, we cannot claim that condition (3.2) is equivalent to the Ehrhard-type inequality (3.4), since the proof of Theorem 3.1 involved an application of the Cauchy-Schwartz inequality (3.6). Consequently, we pose this as a question:

Question 3.4 (Gaussian Neumann-to-Dirichlet Operator on Convex Domains). *Does (3.2) hold for $\mu = \gamma$ the Gaussian measure with $F(v) = \frac{1}{v} - (\log I_\gamma)'(v)$?*

Condition (3.2), together with the Neumann problem to which it applies, is:

Given a compact convex body $K \in C_+^2$ and $f \in C^{1,\alpha}(\partial K)$, let us now solve the following Neumann Laplace equation:

$$Lu \equiv \frac{1}{\mu(K)} \int_{\partial K} f d\mu_{\partial K} \text{ on } K, \quad u_\nu = f \text{ on } \partial K. \quad (3.1)$$

Since the compatibility condition $\int_K Lu d\mu = \int_{\partial K} f d\mu_{\partial K}$ is satisfied, it is known (e.g. [15]) that a solution $u \in C^{2,\alpha}(K)$ exists (and is unique up to an additive constant). The operator mapping $f \mapsto u$ is called the Neumann-to-Dirichlet operator.

Theorem 3.1. *Assume that there exists a function $F : \mathbb{R}_+ \rightarrow \mathbb{R}$ so that for all K , f and u as above:*

$$F(\mu(K)) \left(\int_{\partial K} f d\mu_{\partial K} \right)^2 \leq \int_{\partial K} (\langle \nabla_{\partial K} f, \nabla_{\partial K} u \rangle + u_{\nu,\nu} f) d\mu_{\partial K}. \quad (3.2)$$

For the standard Gaussian, Lemma 3.2 of the source identifies the right-hand side of (3.2) with

$$\mathcal{E}_K(u) := \int_K (\|\nabla^2 u\|_{\text{HS}}^2 + |\nabla u|^2) d\gamma_2. \quad (1)$$

Thus, if $\mathcal{L} = \Delta - \langle x, \nabla \rangle$, $\mathcal{L}u = 1$, and $f = u_\nu$, condition (3.2) requires

$$\mathcal{E}_K(u) \geq F(\gamma_2(K))\gamma_2(K)^2. \quad (2)$$

Proof Intuition

The halfspace of Gaussian measure $1/2$ is an equality case. Its one-dimensional solution has derivative $w = \Phi/\varphi$. The first genuinely curved tangential perturbation is the second Hermite mode $\rho(y) = (1 - y^2)/\sqrt{2}$. Differentiation raises the positive one-dimensional Ornstein–Uhlenbeck eigenvalue: $\mathcal{L}_x w' = 2w'$, while $\mathcal{L}_y \rho = -2\rho$. Hence $w'(x)\rho(y)$ is $\mathcal{L}$ -harmonic and can be added without changing the constant source $\mathcal{L}u = 1$.

Optimizing this correction on a volume-preserving parabolic deformation of the halfspace produces the strictly negative coefficient $-(4 - \pi)^2/(16\pi)$. The parabola itself is unbounded, but large ellipses converge to it in both Gaussian volume and the energy (1). The strict deficit therefore survives on a finite smooth strictly convex ellipse, which satisfies all hypotheses in the source question.

Counterexample Theorem

Theorem 1. *There exist a compact C^∞ strictly convex body $K \subset \mathbb{R}^2$, a function $u \in C^\infty(\overline{K})$ satisfying $\mathcal{L}u = 1$ in K , and the smooth Neumann datum $f = u_\nu$ on ∂K , such that*

$$\gamma_2(K) = \frac{1}{2}, \quad \mathcal{E}_K(u) < \frac{1}{2} = F\left(\frac{1}{2}\right) \left(\int_{\partial K} f d\gamma_{\partial K} \right)^2.$$

Consequently condition (3.2) in the source paper is false for the standard Gaussian measure, and Question 3.4 has a negative answer.

Proof

Write

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad \Phi(x) = \int_{-\infty}^x \varphi(t) dt, \quad w(x) = \frac{\Phi(x)}{\varphi(x)}.$$

Then

$$w' = 1 + xw, \quad \mathcal{L}_x w = w, \quad \mathcal{L}_x w' = 2w'. \quad (3)$$

Choose a smooth primitive W of w . The first identity gives

$$\mathcal{L}_x W = W'' - xW' = w' - xw = 1. \quad (4)$$

Let

$$\rho(y) = \frac{1 - y^2}{\sqrt{2}}.$$

For a standard one-dimensional Gaussian variable Y ,

$$\mathcal{L}_y \rho = -2\rho, \quad \mathbb{E}\rho(Y) = 0, \quad \mathbb{E}\rho(Y)^2 = 1, \quad \mathbb{E}\rho'(Y)^2 = 2. \quad (5)$$

Put $a = w'$ and

$$u_{\varepsilon,A}(x, y) = W(x) + \varepsilon Aa(x)\rho(y). \quad (6)$$

Equations (3)–(5) imply

$$\mathcal{L}u_{\varepsilon,A} = 1 \quad (7)$$

on all of $\mathbb{R}^2$.

The volume-preserving parabolic model

For ε near zero, let

$$P_\varepsilon = \{(x, y) : x < c(\varepsilon) + \varepsilon\rho(y)\},$$

where $c(\varepsilon)$ is chosen so that $\gamma_2(P_\varepsilon) = 1/2$. Such a unique smooth $c(\varepsilon)$ exists by the implicit function theorem applied to

$$V(c, \varepsilon) = \mathbb{E} \Phi(c + \varepsilon\rho(Y)).$$

At $(c, \varepsilon) = (0, 0)$, equations (5) and $\varphi'(0) = 0$ give

$$V_c = \varphi(0) > 0, \quad V_\varepsilon = 0, \quad V_{\varepsilon\varepsilon} = 0.$$

It follows that

$$c(0) = c'(0) = c''(0) = 0, \quad c(\varepsilon) = o(\varepsilon^2). \quad (8)$$

Since $\rho'' = -\sqrt{2}$, the graphing function is strictly concave when $\varepsilon > 0$, and P_ε is convex.

We now compute the energy through order ε^2 . Set

$$q_0 = w'^2 + w^2, \quad B = w'a'' + wa'.$$

Expanding the quadratic energy density of (6), then expanding the upper integration limit $c(\varepsilon) + \varepsilon\rho(y)$, gives

$$\mathcal{E}_{P_\varepsilon}(u_{\varepsilon,A}) = \mathcal{E}_{P_0}(W) + \varepsilon^2(C_0 + 2AL + A^2D) + o(\varepsilon^2), \quad (9)$$

where

$$C_0 = \frac{1}{2}\varphi(0)q'_0(0)\mathbb{E}\rho^2, \quad (10)$$

$$L = \varphi(0)B(0)\mathbb{E}\rho^2, \quad (11)$$

$$D = \int_{\{x < 0\}} \left(\|\nabla^2(a\rho)\|_{\text{HS}}^2 + |\nabla(a\rho)|^2 \right) d\gamma_2. \quad (12)$$

For completeness, the terms of order ε vanish because $\mathbb{E}\rho = 0$. The base-density boundary shift contributes $\frac{1}{2}\varphi(0)q'_0(0)\mathbb{E}\rho^2$; the cross-energy boundary shift contributes $2AL$; the interior quadratic term is A^2D ; and (8) shows that the volume correction contributes only $o(\varepsilon^2)$. This proves (9) directly by Taylor's formula and Gaussian domination.

Put $\alpha = w(0) = \sqrt{\pi/2}$. Successive differentiation of $w' = 1 + xw$ gives

$$w(0) = \alpha, \quad w'(0) = 1, \quad w''(0) = \alpha, \quad w'''(0) = 2, \quad \varphi(0)\alpha = \frac{1}{2}. \quad (13)$$

Because $\mathcal{L}_x w = w$, one-dimensional Gaussian integration by parts yields

$$\mathcal{E}_{P_0}(W) = \int_{-\infty}^0 (w'^2 + w^2)\varphi dx = \varphi(0)w(0)w'(0) = \frac{1}{2}. \quad (14)$$

Equations (10), (11), and (13) give

$$C_0 = 1, \quad L = \varphi(0)(2 + \alpha^2) = \frac{\pi + 4}{2\sqrt{2\pi}}. \quad (15)$$

It remains to evaluate D . Let $\psi = a(x)\rho(y)$. Since $\mathcal{L}\psi = 0$, differentiating shows that $\mathcal{L}(\partial_i\psi) = \partial_i\psi$. Gaussian integration by parts on the halfspace therefore gives

$$\begin{aligned} D &= \varphi(0) \mathbb{E} [\nabla\psi(0, Y) \cdot \partial_x \nabla\psi(0, Y)] \\ &= \varphi(0) \left(a'(0)a''(0)\mathbb{E}\rho^2 + a(0)a'(0)\mathbb{E}(\rho')^2 \right). \end{aligned}$$

Here $a = w'$, so (5) and (13) yield

$$D = \varphi(0)(2\alpha + 2\alpha) = 2. \quad (16)$$

The quadratic coefficient in (9) is minimized at

$$A_* = -\frac{L}{D} = -\frac{\pi + 4}{4\sqrt{2\pi}}. \quad (17)$$

Using (14), (15), and (16), we obtain the exact expansion

$$\mathcal{E}_{P_\varepsilon}(u_{\varepsilon, A_*}) = \frac{1}{2} - \frac{(4 - \pi)^2}{16\pi} \varepsilon^2 + o(\varepsilon^2). \quad (18)$$

In particular, this energy is strictly smaller than $1/2$ for every sufficiently small positive ε .

Compactification by ellipses

Fix one such $\varepsilon > 0$. For $R > 0$ and $c \in \mathbb{R}$, define the ellipse

$$K_{\varepsilon, R, c} = \left\{ (x, y) : \frac{(x - c - \varepsilon/\sqrt{2} + R)^2}{R^2} + \frac{\sqrt{2}\varepsilon}{R} y^2 \leq 1 \right\}. \quad (19)$$

Its right boundary is

$$b_{\varepsilon, R, c}(y) = c + \frac{\varepsilon}{\sqrt{2}} - R + R\sqrt{1 - \frac{\sqrt{2}\varepsilon}{R} y^2}.$$

For every fixed y ,

$$b_{\varepsilon, R, c}(y) \longrightarrow c + \frac{\varepsilon}{\sqrt{2}} - \frac{\varepsilon}{\sqrt{2}} y^2 = c + \varepsilon \rho(y), \quad (20)$$

while the left boundary tends to $-\infty$.

For every sufficiently large R , translate in the x direction and choose a local normalizing parameter $c = c_{\varepsilon, R}$ near $c(\varepsilon)$ such that $\gamma_2(K_{\varepsilon, R, c_{\varepsilon, R}}) = 1/2$. Indeed, on a fixed neighborhood of $c(\varepsilon)$ the volume and its c -derivative converge to $V(c, \varepsilon)$ and $V_c(c, \varepsilon) > 0$, respectively. The implicit function theorem therefore supplies a unique root in that neighborhood for large R , and

$$c_{\varepsilon, R} \longrightarrow c(\varepsilon). \quad (21)$$

The same dominated-convergence argument applied to the energy density of the fixed smooth function u_{ε, A_*} gives

$$\mathcal{E}_{K_{\varepsilon, R, c_{\varepsilon, R}}}(u_{\varepsilon, A_*}) \longrightarrow \mathcal{E}_{P_\varepsilon}(u_{\varepsilon, A_*}) < \frac{1}{2}. \quad (22)$$

To justify domination explicitly, (20) gives a uniform upper bound on x ; as $x \rightarrow -\infty$, the standard Mills-ratio identities from $w' = 1 + xw$ give polynomial decay of w, w', w'', w''' ; and the y dependence is polynomial times the Gaussian density. Thus the energy has an integrable majorant independent of large R .

Choose a finite large R for which the strict inequality in (22) holds, and write $K = K_{\varepsilon, R, c_{\varepsilon, R}}$. This K is compact, C^∞ , and has everywhere positive curvature. Let $u = u_{\varepsilon, A_*}$ and $f = u_\nu$. By (7) and Gaussian integration by parts,

$$\int_{\partial K} f d\gamma_{\partial K} = \int_K \mathcal{L}u d\gamma_2 = \gamma_2(K) = \frac{1}{2}. \quad (23)$$

Finally, $\Phi^{-1}(1/2) = 0$, so $(\log I_\gamma)'(1/2) = 0$ and $F(1/2) = 2$. Equations (22) and (23) show

$$\mathcal{E}_K(u) < \frac{1}{2} = 2 \left(\frac{1}{2} \right)^2 = F(\gamma_2(K)) \left(\int_{\partial K} f d\gamma_{\partial K} \right)^2.$$

This is the required violation of condition (3.2), proving Theorem 1. $\square$

Verification Notes

The proof is analytic and does not depend on numerical quadrature. The included script `code/check_second_variation_and_ellipses.py` checks the identity

$$1 - \frac{1}{2} \left(\frac{\pi + 4}{2\sqrt{2\pi}} \right)^2 = -\frac{(4 - \pi)^2}{16\pi}$$

to floating-point precision and independently integrates finite ellipses. For example, at $\varepsilon = 0.03$ and $R = 100$, after translating to numerical Gaussian volume 0.5, it reports energy approximately 0.49999660148. These computations are sanity checks only.

Initial reconnaissance optimized every interval on a 65-point endpoint grid (2016 intervals), scanned centered Gaussian balls in dimensions 2 through 10, and tested product halfspace models. No violation appeared because the negative mode requires both curvature and a nontrivial tangential direction.

Novelty Check and Limitations

Before promotion, the cheap run indexes were searched for arXiv:1601.02925, the exact source title, “Gaussian Neumann-to-Dirichlet”, and the fixed-source Hessian-energy formulation. No prior solution or counterexample packet for this question was present.

Web and arXiv searches on 2026-08-09 used the exact question title, the source authors and arXiv id, “Gaussian Neumann-to-Dirichlet convex domains”, and “Gaussian Hessian energy Neumann convex”. They found the source question but no later paper claiming to resolve it or containing this parabolic counterexample. Kai-Wen Yang’s 2026 paper [2] was inspected because it uses raywise localization of Gaussian Neumann Hessian energy. Its argument selects a variable source $\mathcal{L}u = g$ for a different Brunn–Minkowski problem; it does not answer the present fixed-constant-source condition.

This was a bounded rather than exhaustive novelty search. The packet claims a likely valid candidate counterexample, not certified publication novelty.

Human Review Recommendation

The recommended review focus is narrow: check the Taylor expansion (9), especially the cross-energy boundary shift, and check the dominated-convergence passage (21)–(22). All remaining steps are explicit Ornstein–Uhlenbeck identities and Gaussian integration by parts.

References

- [1] A. V. Kolesnikov and E. Milman, *Sharp Poincaré-type inequality for the Gaussian measure on the boundary of convex sets*, Geometric Aspects of Functional Analysis, Lecture Notes in Mathematics 2169, 221–234, 2017; arXiv:1601.02925.
- [2] K.-W. Yang, *The Brunn–Minkowski inequality for the Gaussian measure*, arXiv:2608.05390, 2026.